# MABEL Arm Inertial-Parameter Calibration

Identifying link mass and centre of mass of the 7-DOF right arm for inverse dynamics and gravity compensation

MABEL calibration

June 16, 2026

This package (`mabel_sysid`) identifies the *rigid-body* inertial parameters — link **mass** $m_i$ and **centre of mass** $c_i$ — of the MABEL right arm, for use in the inverse-dynamics and gravity-compensation feed-forward terms. It deliberately treats the joints as ideal (zero friction and damping); those neglected motor/gearbox effects are identified separately by the companion `mabel_actid` package (see `ACTUATOR_CALIBRATION.pdf`). The whole pipeline is *proven in simulation first*: the known MuJoCo masses are taken as ground truth, scrambled, and recovered.

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# 1 Method

## 1.1 Linear-in-parameters dynamics

The equation of motion of the arm is

$$M(q)\,\ddot{q} + C(q,\dot{q})\,\dot{q} + g(q) = \tau. \tag{1}$$

Although $M, C, g$ are nonlinear in the joint angles $q$, each link contributes its mass $m_i$, first moment $\varphi_i = m_i c_i \in \mathbb{R}^3$, and inertia tensor $I_i \in \mathbb{R}^{3\times3}$, and the dynamics are *linear* in this 10-parameter-per-link vector $\pi$:

$$\tau = Y(q,\dot{q},\ddot{q})\,\pi, \tag{2}$$

with $Y$ the *regressor*. Identification is therefore a linear regression: stack the rows $Y(q_k,\dot{q}_k,\ddot{q}_k)$ measured along a trajectory and solve a least-squares problem for $\pi$. Code: `src/mabel_sysid/regressor.py`, `identify.py`.

### 1.2 Gravity needs only the first moments

Gravity compensation requires only $g(q)$, which depends solely on the first moments $\varphi_i = m_i c_i$ (not the full inertia, and not $m_i$ and $c_i$ separately):

$$g(q) = \sum_i \frac{\partial}{\partial q}\big( - m_i\, \mathbf{g}^\top r_i(q)\big) = Y_g(q)\, \varphi, \tag{3}$$

so a set of *static holding poses* — where $\dot{q} = \ddot{q} = 0$ and the measured torque is exactly $g(q)$ — is enough to fit the gravity model. This is the cheapest and most robust experiment.

### 1.3 Two estimators

- **Method B (linear).** Closed-form least squares on the analytic gravity regressor $Y_g$ over the static poses; returns the first moments $\varphi$ directly and exposes observability through the singular spectrum of $Y_g$.
- **Method A (nonlinear).** A MuJoCo-in-the-loop optimiser that adjusts $(m_i, c_i)$ to match the full inverse-dynamics torque over static poses plus a Fourier excitation trajectory; this is what separates $m_i$ from $c_i$ (mass enters only through the acceleration term $M(q)\ddot{q}$).

## 2 Identifiability

A fundamental limit governs how much can be recovered. Static (gravity) data sees only first moments $m_i c_i$, so a link's own mass — which barely moves its frame origin — is *unobservable* from gravity alone. Separating mass from COM needs acceleration. In general only certain fixed combinations, the *base parameters*, are identifiable; the rest lie in the regressor's null space.

For the MABEL arm the gravity regressor has $7 \times 4 = 28$ parameters but only **14 observable directions** (Table 1): the singular spectrum drops by ∼10 orders of magnitude after the 14th value. The controller only needs those identifiable combinations — and they are recovered to the noise floor — so gravity compensation is excellent even though some individual link masses remain partly indeterminate.

## 3 Simulation validation and results

The model is built from the full robot (`tools/extract_arm_model.py` slices the right-arm chain out of `mabel_full.xml`, fixes it to the world with a small base tilt so gravity excites every joint, and zeroes joint friction/damping). The true masses are read, scrambled by ±30% (with COMs shifted up to ±2 cm) to form a "CAD prior", and then identified from 2250 noisy samples ($\sigma_\tau = 0.05$ N m).

The torque predictions improve by roughly 50×, well below the 0.05 N m sensor-noise floor. Per-link mass (Table 2) is only partially corrected — the expected identifiability limit — while the controller-relevant first moments and the torque fit are essentially exact.

## 4 Applying to the real robot

The procedure is identical except for the data source. Replace the synthetic dataset with recorded $(q, \dot{q}, \ddot{q}, \tau)$: command static poses spanning the workspace and log the steady holding torque ($i_q \cdot K_t$, gravity-consistent) for gravity comp; add a smooth Fourier excitation and filtered differentiation for the full inverse-dynamics fit. Use the CAD values as the prior, validate on held-out poses, and load

Table 1: Gravity-compensation and inverse-dynamics validation. "before" uses the scrambled prior; "after" uses the identified parameters, on held-out poses.

| metric | before (prior) | after (identified) |
|---|---|---|
| gravity-torque RMS (N m) | 0.2024 | **0.0037** |
| inverse-dynamics torque RMS (N m) | 0.4038 | **0.0038** |
| COM mean abs. error (mm) | — | 11.4 |
| per-link mass MAE (kg) | — | 0.117 |
| observable directions / params | 14 / 28 | |

Table 2: Per-link mass: ground truth vs. identified (nonlinear and linear estimators). Distal links recover well; large proximal links stay partly in the null space.

| link | true (kg) | id. nonlinear (kg) | id. linear (kg) |
|---|---|---|---|
| `r_shoulder_1` | 0.731 | 0.592 | 0.571 |
| `r_shoulder_2` | 0.698 | 0.672 | 0.712 |
| `r_shoulder_3` | 1.466 | 1.904 | 1.893 |
| `r_elbow` | 0.942 | 0.978 | 1.006 |
| `r_forearm` | 0.837 | 0.698 | 0.644 |
| `r_wrist` | 0.309 | 0.338 | 0.280 |
| `r_hand_mount` | 0.318 | 0.332 | 0.395 |

the result into the controller's feed-forward term $\tau_{\text{ff}} = \tau_{\text{rb}}(q, \dot{q}, \ddot{q})$, on top of which the `mabel_actid` friction term is added. Account for the joint friction/damping that is zeroed here (identify it, or include it in $\tau$), and make sure the mount orientation relative to gravity is correct.

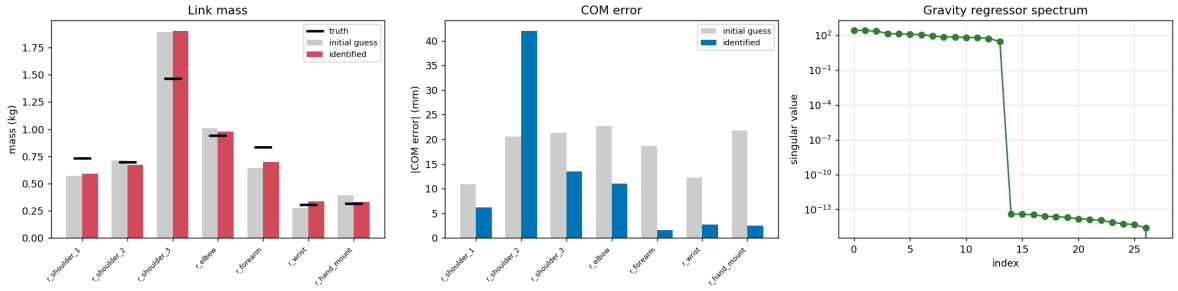



Figure 1: Left: per-link mass — identified (red) vs. truth (black) vs. scrambled start (grey). Middle: COM error, identified vs. prior. Right: the gravity regressor singular spectrum — 14 large values then a cliff to $\sim 10^{-13}$, which *is* the identifiability limit.

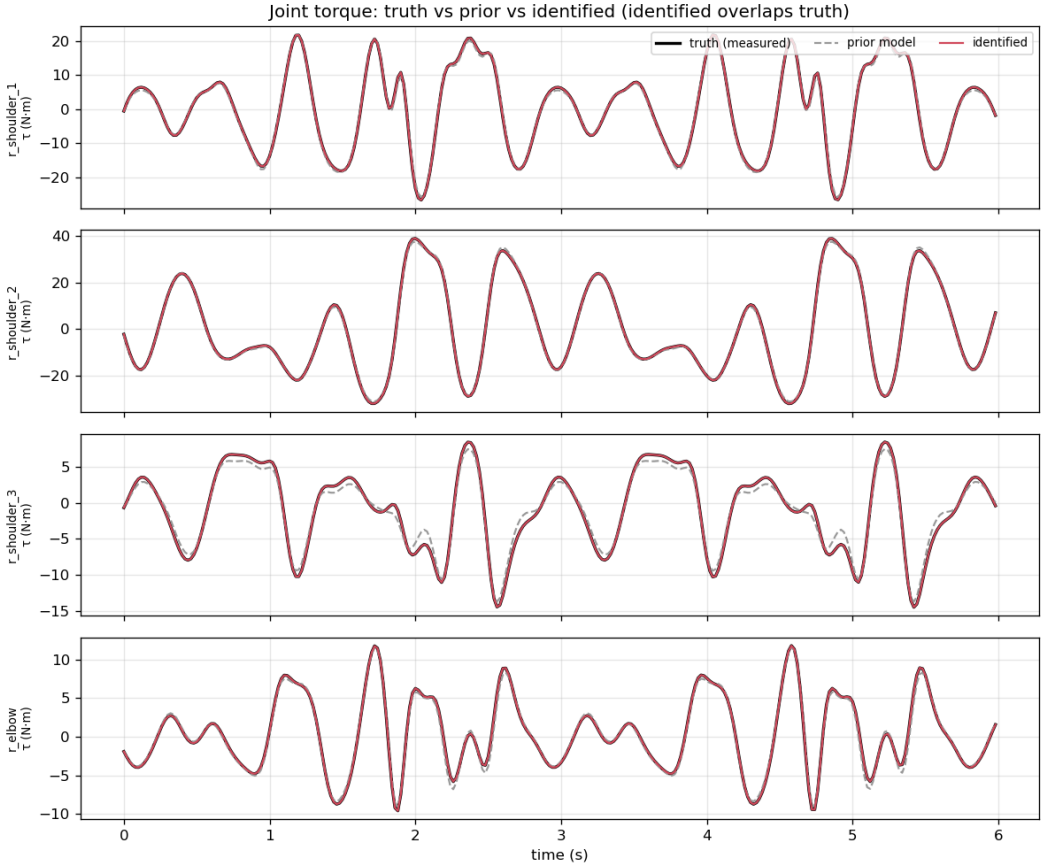



Figure 2: Joint torque along the excitation trajectory: truth vs. prior vs. identified. The identified curve overlaps the truth.